

EPC-3D-Diff: Equivariant Physics Consistent Conditional 3D Latent Diffusion for CBCT to CT Synthesis

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Background and Objective

- **Problem:** Cone-beam CT (CBCT) routinely acquired during radiotherapy for patient positioning and image guidance. However, scatter, beam hardening, detector effects, and reconstruction artifacts substantially degrade image quality $\Rightarrow$ limitations to use CBCT for dose calculation and adaptive radiotherapy.
- **Solution:** A physics consistent 3D latent diffusion framework that introduces projection domain rotational equivariance to improve HU fidelity and robustness across scanner and domain shifts.
- **Challenge:** Can we synthesize CT quality images from CBCT while preserving anatomy, recovering reliable HU values, and consistent with acquisition physics?

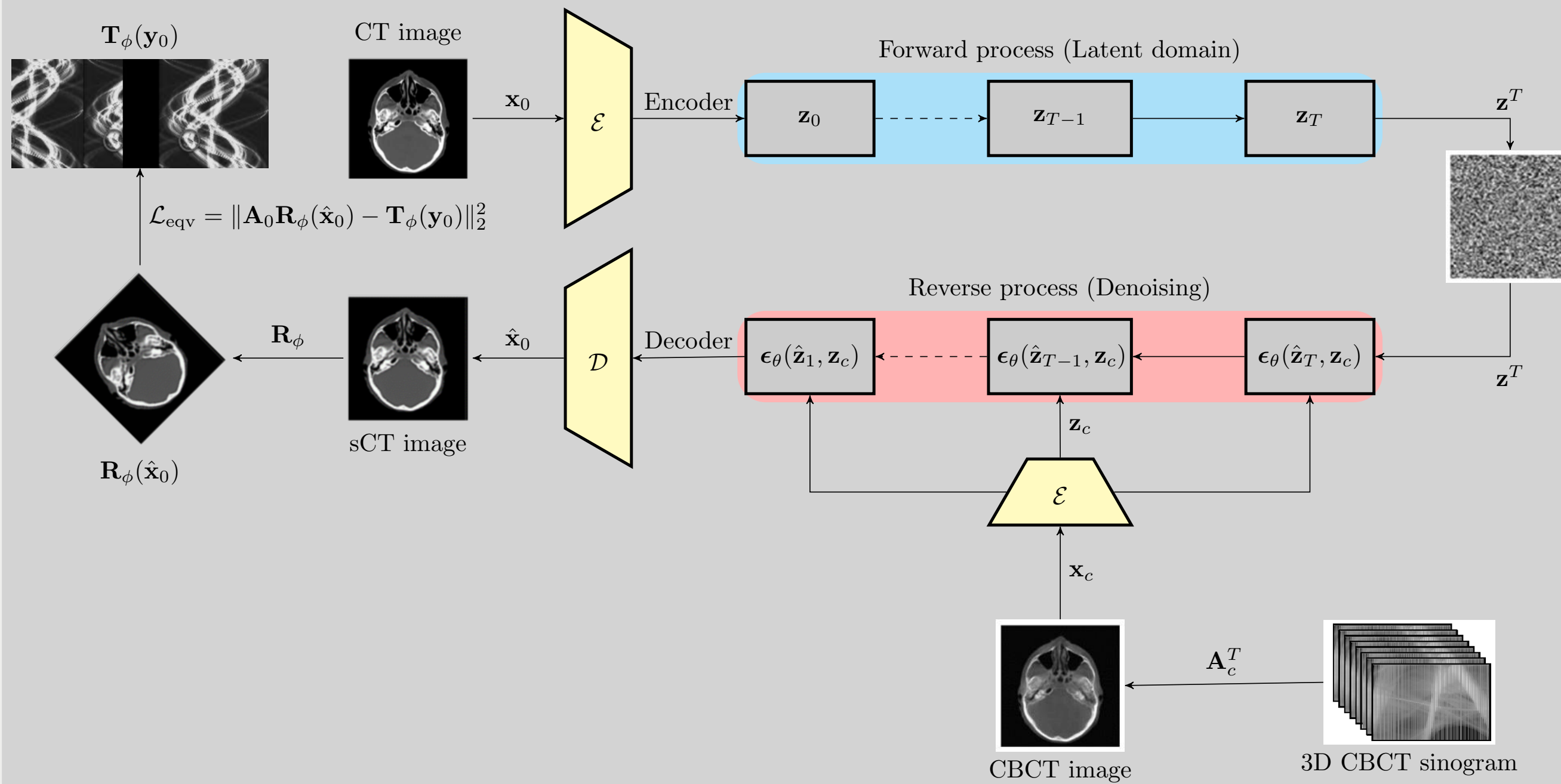
Proposed Framework (EPC-3D-Diff)

Novel framework: **Equivariant Physics Consistent Conditional 3D Latent Diffusion for CBCT to CT Synthesis (EPC-3D-Diff).**

- Conditional Diffusion Model in a 3D latent space:** scalable volumetric CBCT to CT synthesis.
- Equivariant rotational design:** exploit CT operator symmetries to improve robustness under scanner and domain shifts.
- Physics consistent measurement domain constraints:** derived from the CT forward model to discourage physically implausible HU solution.

EPC-3D-Diff with Projection Equivariant Loss

- sCT $\hat{\mathbf{x}}_0$ rotated via $\mathbf{R}_\phi$ to enforce rotational consistency. CT forward operator $\mathbf{A}_0$ and rotation T_ϕ applied in the projection domain for the equivariant loss $\mathcal{L}_{eq}$.



Equivariant Physics Consistent Conditional 3D Latent Diffusion

- **Latent 3D representation to improve computational efficiency:** let $E(\cdot)$ and $D(\cdot)$ Autoencoder (AE) and decoder pair for CT and CBCT volumes $\mathbf{z}_0 = E_\psi(\mathbf{x}_0)$, $\mathbf{z}_c = E_\psi(\mathbf{x}_c)$, $E_\psi: \mathbb{R}^{N \times M \times S} \rightarrow \mathbb{R}^{N' \times M' \times S}$, $N' < N$, $M' < M$
- **Forward diffusion process:** noisy latent vectors $\mathbf{z}_t$, time step t obtained by adding noise progressively: $\mathbf{z}_t = \sqrt{\alpha_t} \mathbf{z}_0 + \sqrt{1 - \alpha_t} \epsilon$, $\epsilon \sim N(0, \mathbf{I})$
- **EPC-3D-Diff Reverse process with guidance:** CBCT measurements $\mathbf{y}_c$ mapped to image domain data $\mathbf{x}_c = \mathbf{A}_c^T \mathbf{y}_c$ using CBCT physics operator $\mathbf{A}_c$. From the predicted noise $\epsilon_\theta(\mathbf{z}_t, t, E_\psi(\mathbf{x}_c))$, the 3D sCT obtained using the AE decoder $\hat{\mathbf{x}}_0 = D_\omega(\hat{\mathbf{z}}_0)$ from latent representation $\hat{\mathbf{z}}_0$. DDPM training objective: $\mathcal{L}_{DDPM}(\theta) = \mathbb{E}_{t, \mathbf{z}_0, \epsilon} \|\epsilon - \epsilon_\theta(\mathbf{z}_t, t, E_\psi(\mathbf{x}_c))\|_2^2$
- **Projection Equivariant Diffusion Consistency:** CBCT imaging rotational symmetry of the source-detector trajectory $\mathbf{A}_0(\mathbf{R}_\phi \hat{\mathbf{x}}_0)(\eta) = \mathbf{A}_0(\mathbf{x})(\eta - \phi)$ synthetic CT measurement generated by rotation $\hat{\mathbf{x}}_0$ using forward CBCT operator $\mathbf{A}_0 \Rightarrow$ match the translated CT measurements $\mathbf{y}_0$ with loss

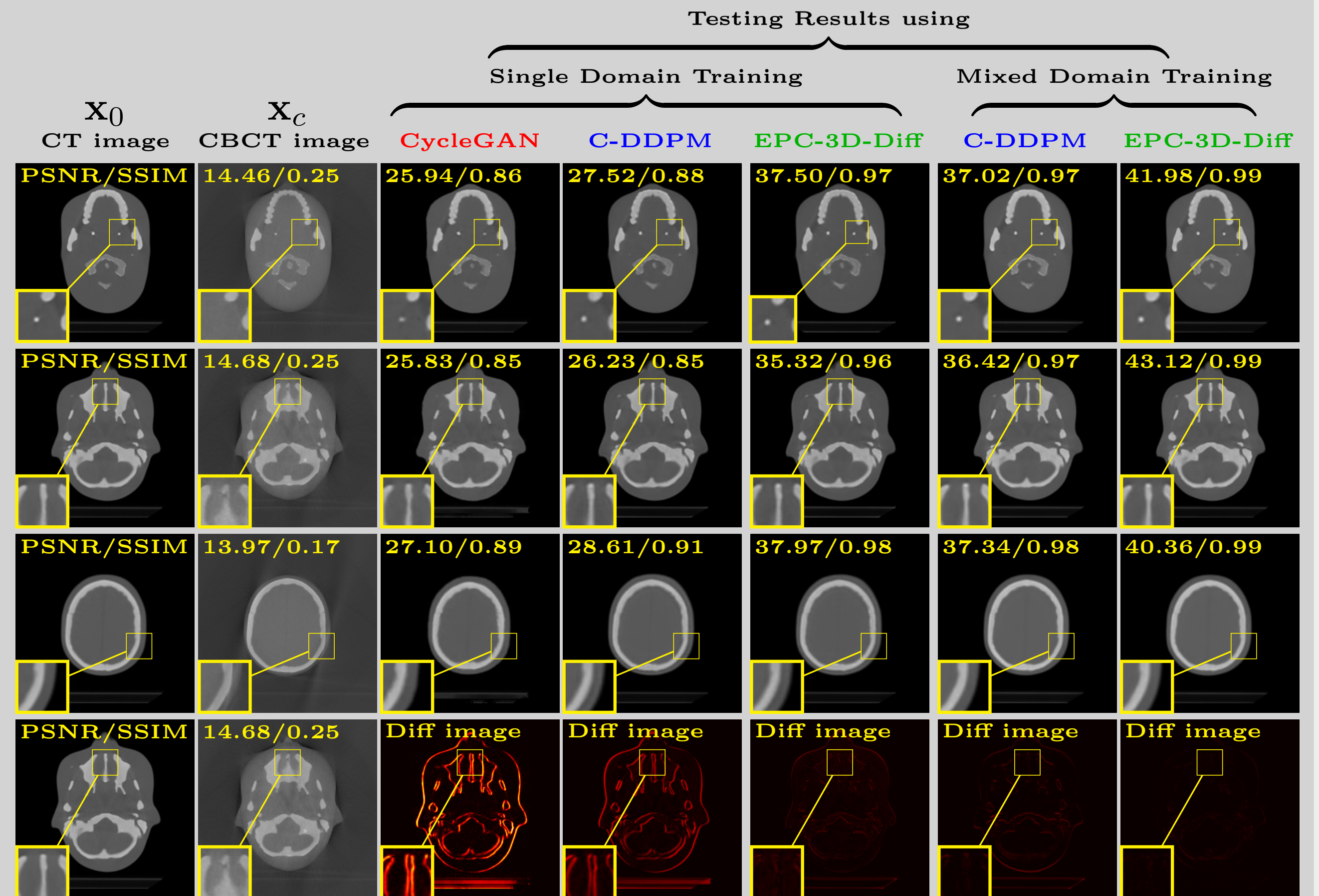
$$\mathcal{L}_{eq} = \sum_{i=1}^{N_\phi} \|\mathbf{T}_{\phi_i}(\mathbf{y}_0) - \mathbf{A}_0 \mathbf{R}_{\phi_i}(\hat{\mathbf{x}}_0)\|_2^2,$$

with N_ϕ the number of angular rotations and $\mathbf{T}_\phi$ the fractional angular shift.

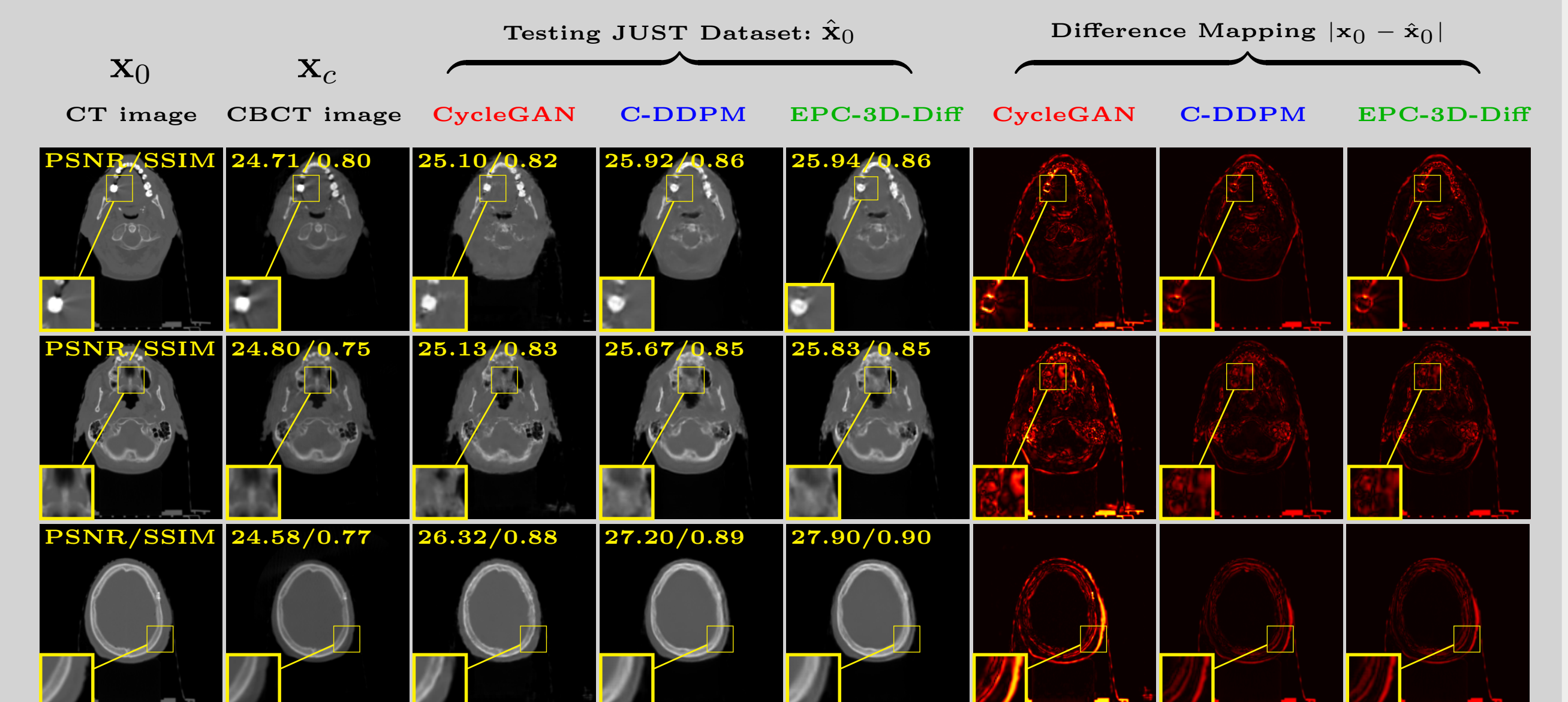
- **EPC-3D-Diff Inference Strategy:** only the CBCT input $\mathbf{x}_c$ is available $\Rightarrow$ sampling from the learned conditional prior $\mathbf{x} \sim p_\theta(\mathbf{x} | \mathbf{x}_c)$ using deterministic DDIM updates $\Rightarrow$ inference remains operator free and computationally efficient.

Results

- Dataset: (JUST): 11 cases training (631 slices), 3 testing (173 slices). (NHW): phantom 8 cases training (736 slices), 2 testing (184 slices).
- Results for single-domain (NWH) and mixed-domain (NWH+JUST) training.



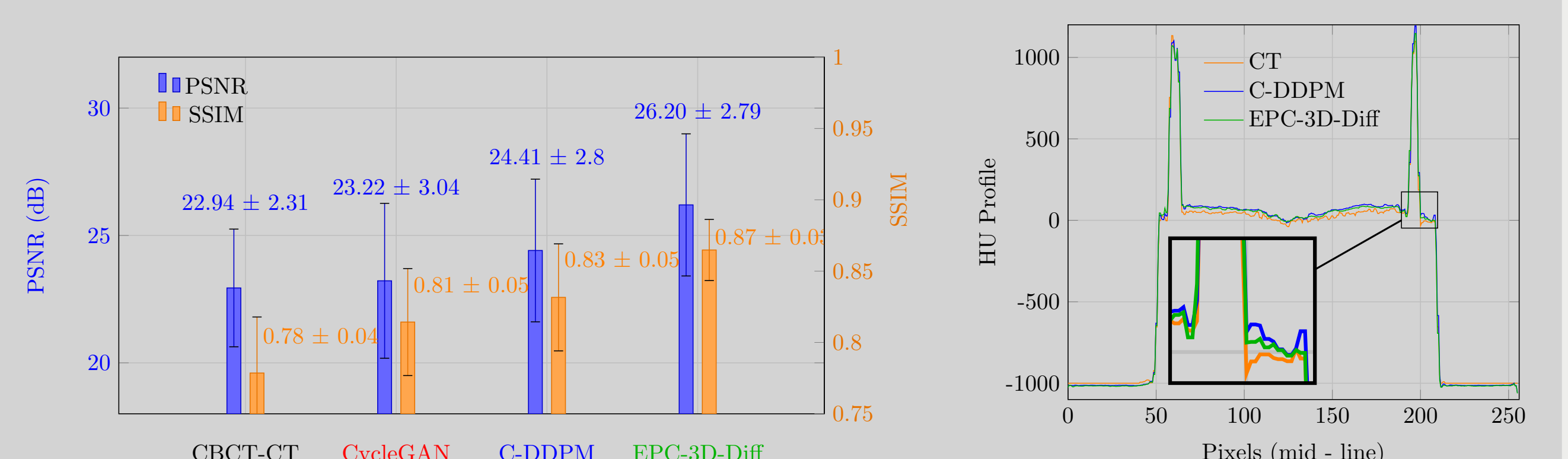
- Qualitative testing comparison on JUST clinical test set.



- Quantitative results (mean $\pm$ std) on the NWH phantom test set on single and multiple domain training for CycleGAN, C-DDPM, and EPC-3D-Diff.

	Method	PSNR (dB)	SSIM	MSE (dB)
	CBCT - CT	14.41 $\pm$ 0.351	0.26 $\pm$ 0.07	-33.14 $\pm$ 0.37
Single Domain Training	CycleGAN	25.85 $\pm$ 7.82	0.82 $\pm$ 0.09	-62.38 $\pm$ 4.78
	C-DDPM	30.77 $\pm$ 4.31	0.91 $\pm$ 0.05	-66.53 $\pm$ 3.53
	EPC-3D-Diff	30.99 $\pm$ 4.21	0.92 $\pm$ 0.05	-66.69 $\pm$ 5.38
Multiple Domain Training	CycleGAN	30.50 $\pm$ 7.27	0.88 $\pm$ 0.07	-62.35 $\pm$ 4.65
	C-DDPM	31.00 $\pm$ 4.48	0.93 $\pm$ 0.05	-66.69 $\pm$ 3.66
	EPC-3D-Diff	38.44 $\pm$ 2.28	0.99 $\pm$ 0.01	-86.86 $\pm$ 3.39

- Quantitative analysis on JUST clinical test set.



References

- [1] Alzahra Altalib et al. "Synthetic CT image generation from CBCT: a systematic review". In: *IEEE Transactions on Radiation and Plasma Medical Sciences* 9.6 (2025), pp. 691-707

Ethical approval JUST clinical study granted by the Research Ethics Committee (Mar2026/191-38).